The Playoff Example (solved using tree diagrams in previous lecture)

Two teams A and B play a play-off series. Whoever wins 3 games becomes champion; there are no ties. How many possible outcomes exist for team A to become the champion?

- The output can be represented as a bit string with up to 5 bits, where each bit corresponds to a game.
- 1 for some bit means that team A won the game. 0 means that team B won the game. There are no ties.
- Total ways for A to win = number of bit strings that contain three 1s: $C(5,3) = 10$.
- Total ways for B to win = number of bit strings that contain three 0s: $C(5,3) = 10$.
- Total ways for series= $10 + 10 = 20$.
- Observation: There cannot be a bit string of length 5 that contains both three 1s and three 0s